

Zhuowei Xu

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Education

Carnegie Mellon University

Master of Science in Mechanical Engineering

Pittsburgh, United States

Sept 2024 – May 2026(Expected)

GPA: 3.9/4.0

Relevant Coursework and Grades: *Mechanics of Manipulation (A), Robot Dynamics and Analysis (A), Machine Learning and Artificial Intelligence for Engineers (A), Numerical Methods in Engineering (A), Modern Control for Robotics (A), Computer Vision (A)*

Shanghai Jiao Tong University

Bachelor of Mechanical Engineering

Shanghai, China

Aug 2020 – Jul 2024

GPA: 3.5/4.0

Relevant Coursework and Grades: *Calculus I (95.2), Introduction to Engineering (96), Theoretical Mechanics (96), Engineering Material (90), Introduction to Robotics (95), Mechanical Dynamics (91)*

Publications

- Yongzhou Long, Zhuang Zhang, **Zhuowei Xu**, Enlin Gu, Qiujie Lu, Hao Wang, Genliang Chen. “Lightweight and Powerful Vacuum-Driven Gripper with Bioinspired Elastic Spine.” [IEEE Robotics and Automation Letters. <https://ieeexplore.ieee.org/document/10287554>].
- **Zhuowei Xu**, Zilin Si, Kevin Lee Zhang, Moonyoung Lee, Shashwat Singh, Oliver Kroemer, Zeynep Temel. “A Low-Cost Tactile Fingertip Design for Dexterous Robotic Hands.” [ICRA 2025 Workshop ”Handy Moves: Dexterity in Multi-Fingered Hands”. <https://openreview.net/forum?id=5kj8uifi2w>].
- **Zhuowei Xu**, Zilin Si, Kevin Lee Zhang, Oliver Kroemer, Zeynep Temel. “A Multi-modal Tactile Fingertip Design for Robotic Hands to Enhance Dexterous Manipulation.” [Under review. <https://arxiv.org/pdf/2510.05382>].

Research Experience

Object-Centric Key-Frame Extraction for Long-Horizon Tactile Decision-Making

Sept 2025 – Apr 2026

Advisors: Zeynep Temel, Oliver Kroemer, Robotics Institute, Carnegie Mellon University

- Developed a hierarchical tactile decision-making framework that decomposes manipulation into object-centric exploration and execution, enabling exploratory skills to infer action-relevant properties including stiffness, smoothness, weight, and granularity.
- Proposed a top- k tactile key-frame extraction method to identify informative moments from long-horizon tactile interactions and convert them into compact object-centric representations for property-conditioned downstream manipulation.

Tactile-Augmented Imitation Learning Policy for Fine-Grained Manipulation

May 2025 – Apr 2026

Advisors: Zeynep Temel, Oliver Kroemer, Robotics Institute, Carnegie Mellon University

- Developed a tactile-augmented imitation learning framework for dexterous manipulation, integrating vision, force, and vibrotactile sensing.
- Built a kinematic-twin-based teleoperation system for whole-body control of the robotic hand and Franka arm, with a MuJoCo virtual twin for state monitoring and control command visualization.
- Demonstrated automated grasping of fragile and contact-sensitive objects, where force-aware policies achieved high success rate on potato chips pinching.

Multi-modal Tactile Fingertip Design for Robotic Hands

Oct 2024 – Sept 2025

Advisors: Zeynep Temel, Oliver Kroemer, Robotics Institute, Carnegie Mellon University

- Developed a low-cost, easy-to-make, adaptable, and compact multi-modal fingertip integrating strain gauges for force sensing and a contact microphone for vibrotactile sensing; capable of sensing 2D planar force (0–5 N) and classifying different materials.
- Evaluate across three manipulation tasks with different visual occlusion levels, showing that tactile sensing overcomes occlusion, runs efficiently on lightweight hardware, and captures object properties unreachable to vision based models.

Bionic Vacuum-Driven Gripper with a Elastic Spine

Sept 2022 – Nov 2023

Advisors: Genliang Chen, School of Mechanical Engineering, Shanghai Jiao Tong University

- Developed a lightweight and powerful vacuum-driven actuator inspired by snakes’ spine and ribs; achieved response time of 320 ms and maximum grasp force > 50 N.

- Designed and 3D-modeled the spine-rib structure and the actuator; devised an integral forming technique by 3D printing PLA ribs on TPU skin, reducing single-chamber manufacture time to < 40 minutes; built a three-fingered gripper and a wireless handheld mobile gripper.

Course Project

Model-Based Trajectory Control and Planning for a Whiteboard Robot

Feb 2025 – May 2025

Leading the software development

- Built real-time control and dynamics for a whiteboard-erasing robot; build visual tracking system for target track.
- Implemented a 2D simulation environment and benchmarked PID, MPC, and TVLQR under noise; validated on hardware, with MPC delivering the best tracking.

Design and Simulation of Six-DoF Robot with High Load-to-Weight Ratio

Mar 2023 – Jun 2023

Leading the simulation system development

- Designed six-DoF serial robots with a high load-to-weight ratio; implemented control, path planning, and obstacle avoidance using RRT.
- Conducted Simulink-SolidWorks co-simulation; developed a packaged Simscape simulation platform for various environments and loads; implemented an inverted pendulum demo controlled by double-loop PID; selected motors and verified moment requirements.

Professional Experience

Underactuated Hand with Dexterous Manipulation

Jun 2023 – Sept 2023

Research Intern | Fraunhofer Project Center for Smart Manufacturing at SJTU

- Developed an underactuated finger driven by pouch and face actuators, achieving a lifting capacity of 200 g at 0.1 MPa.
- Improved actuation performance through exoskeleton-based stiffness enhancement and variable-stiffness joint design.

TCP Calibration for a Galvo Mirror-Robot Co-Work System

Nov 2023 – May 2024

Research Intern | Shanghai BOCHU Electronic Technology

- Built a galvo mirror-robot co-work system that extends planar galvo processing to 3D surface operations using a 6-DoF robotic arm.
- Developed a kinematic calibration model with error compensation, identifying system parameters from planar images to establish an accurate workspace mapping.

Honors and Awards

- Outstanding Undergraduate Scholarship of SJTU, 2021
- Second Prize, National Physics Competition for College Students in China, 2021
- Second Prize, RoBoMaster “YunHan Cup” Competition of SJTU, 2021

Leadership and Responsibilities

- Graduate Mentor of the CMU SURA (Summer Undergraduate Research Apprenticeship) Project (2025–2026)
- Co-director of the publicity department of the student union (2021–2022)
- Volunteer for freshman registration (Sept 2021, Sept 2022)
- Volunteer teaching during vacations (2020–2023)

Skills

Programming: C++, Python, MATLAB/Simulink, Julia

Hardware Tools: CAD Design, PCB Design, 3D Printing, Robot Operation System(ROS), Arduino